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Mimir Metals — Tooling Lists

All 6 wire families for MM-14 (JBF-30B4SUL)

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Naming convention:

MM-[diameter code]-[type][station]

Types: D=Die, P=Punch, PP=Punch Pin, SP=Spacer, KO=Knockout Pin, F=Finger

Example: MM-250-D1 = 1/4" family, Die Station 1

Each family uses 4 stations:

ST1 — Upset / point initiation

ST2 — Cone / point forming

ST3 — Head form / point finish

ST4 — Final head coin / size

2 fingers (F1, F2) transfer parts between stations.

""

tooling_lists = {

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# _____
# 1/4" WELD STUD FAMILY (wire dia 0.250)
# _____
"MM-250-WS": {
  "family": "1/4\" Weld Stud",
  "wire_dia_in": 0.250,
  "machine": "MM-14",
  "station_1": {
    "description": "Upset / Point Initiation",
    "die": "MM-250-D1",
    "punch": "MM-250-P1",
    "punch_pin": "MM-250-PP1",
    "spacer": "MM-250-SP1",
    "ko_pin": "MM-250-KO1",
  },
  "station_2": {
    "description": "Cone / Point Forming",
    "die": "MM-250-D2",
    "punch": "MM-250-P2",
    "punch_pin": "MM-250-PP2",
    "spacer": "MM-250-SP2",
    "ko_pin": "MM-250-KO2",
  },
}
```

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"station_3": {
  "description": "Head Form / Point Finish",
  "die":      "MM-250-D3",
  "punch":    "MM-250-P3",
  "punch_pin": "MM-250-PP3",
  "spacer":   "MM-250-SP3",
  "ko_pin":   "MM-250-KO3",
},
"station_4": {
  "description": "Final Head Coin / Size",
  "die":      "MM-250-D4",
  "punch":    "MM-250-P4",
  "punch_pin": "MM-250-PP4",
  "spacer":   "MM-250-SP4",
  "ko_pin":   "MM-250-KO4",
},
"fingers": {
  "finger_1": "MM-250-F1",
  "finger_2": "MM-250-F2",
},
"notes": (
  "All 4 stations active. "
  "Fingers sensitive on this family — check clamp pressure before run. "
  "ST1 punch pin sets point depth, do not substitute with larger family."
),
},

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# _____
# 5/16" WELD STUD FAMILY (wire dia 0.312)
# _____

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"MM-312-WS": {
  "family":    "5/16\" Weld Stud",
  "wire_dia_in": 0.312,
  "machine":   "MM-14",
  "station_1": {
    "description": "Upset / Point Initiation",
    "die":      "MM-312-D1",
    "punch":    "MM-312-P1",
    "punch_pin": "MM-312-PP1",
    "spacer":   "MM-312-SP1",
    "ko_pin":   "MM-312-KO1",
  },
  "station_2": {
    "description": "Cone / Point Forming",

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    "die":      "MM-312-D2",
    "punch":    "MM-312-P2",
    "punch_pin": "MM-312-PP2",
    "spacer":   "MM-312-SP2",
    "ko_pin":   "MM-312-KO2",
  },
  "station_3": {
    "description": "Head Form / Point Finish",
    "die":      "MM-312-D3",
    "punch":    "MM-312-P3",
    "punch_pin": "MM-312-PP3",
    "spacer":   "MM-312-SP3",
    "ko_pin":   "MM-312-KO3",
  },
  "station_4": {
    "description": "Final Head Coin / Size",
    "die":      "MM-312-D4",
    "punch":    "MM-312-P4",
    "punch_pin": "MM-312-PP4",
    "spacer":   "MM-312-SP4",
    "ko_pin":   "MM-312-KO4",
  },
  "fingers": {
    "finger_1": "MM-312-F1",
    "finger_2": "MM-312-F2",
  },
  "notes": (
    "All 4 stations active. "
    "ST2 die is shared with 1/4\" family on older setups — "
    "confirm MM-312-D2 is loaded, not MM-250-D2."
  ),
},

```

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# _____
# 3/8" WELD STUD FAMILY (wire dia 0.375")
# _____

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"MM-375-WS": {
  "family":    "3/8\" Weld Stud",
  "wire_dia_in": 0.375,
  "machine":   "MM-14",
  "station_1": {
    "description": "Upset / Point Initiation",
    "die":      "MM-375-D1",
    "punch":    "MM-375-P1",
  }
}

```

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    "punch_pin": "MM-375-PP1",
    "spacer": "MM-375-SP1",
    "ko_pin": "MM-375-KO1",
  },
  "station_2": {
    "description": "Cone / Point Forming",
    "die": "MM-375-D2",
    "punch": "MM-375-P2",
    "punch_pin": "MM-375-PP2",
    "spacer": "MM-375-SP2",
    "ko_pin": "MM-375-KO2",
  },
  "station_3": {
    "description": "Head Form / Point Finish",
    "die": "MM-375-D3",
    "punch": "MM-375-P3",
    "punch_pin": "MM-375-PP3",
    "spacer": "MM-375-SP3",
    "ko_pin": "MM-375-KO3",
  },
  "station_4": {
    "description": "Final Head Coin / Size",
    "die": "MM-375-D4",
    "punch": "MM-375-P4",
    "punch_pin": "MM-375-PP4",
    "spacer": "MM-375-SP4",
    "ko_pin": "MM-375-KO4",
  },
  "fingers": {
    "finger_1": "MM-375-F1",
    "finger_2": "MM-375-F2",
  },
  "notes": (
    "All 4 stations active. "
    "3/8\" is the highest volume family — check die wear at every "
    "50K pieces. KO pins at ST3 and ST4 wear faster on this size."
  ),
},

```

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#
# 1/2" WELD STUD FAMILY (wire dia 0.500")
#

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```

"MM-500-WS": {
  "family": "1/2\" Weld Stud",

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```

"wire_dia_in": 0.500,
"machine": "MM-14",
"station_1": {
  "description": "Upset / Point Initiation",
  "die": "MM-500-D1",
  "punch": "MM-500-P1",
  "punch_pin": "MM-500-PP1",
  "spacer": "MM-500-SP1",
  "ko_pin": "MM-500-KO1",
},
"station_2": {
  "description": "Cone / Point Forming",
  "die": "MM-500-D2",
  "punch": "MM-500-P2",
  "punch_pin": "MM-500-PP2",
  "spacer": "MM-500-SP2",
  "ko_pin": "MM-500-KO2",
},
"station_3": {
  "description": "Head Form / Point Finish",
  "die": "MM-500-D3",
  "punch": "MM-500-P3",
  "punch_pin": "MM-500-PP3",
  "spacer": "MM-500-SP3",
  "ko_pin": "MM-500-KO3",
},
"station_4": {
  "description": "Final Head Coin / Size",
  "die": "MM-500-D4",
  "punch": "MM-500-P4",
  "punch_pin": "MM-500-PP4",
  "spacer": "MM-500-SP4",
  "ko_pin": "MM-500-KO4",
},
"fingers": {
  "finger_1": "MM-500-F1",
  "finger_2": "MM-500-F2",
},
"notes": (
  "All 4 stations active. "
  "1/2\" requires higher feed roll pressure than smaller families. "
  "Verify finger timing CAM settings before first cycle — "
  "heavier wire can cause transfer misses at ST2."
),

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},

9/16" WELD STUD FAMILY (wire dia 0.562")

```
"MM-562-WS": {  
  "family":    "9/16\" Weld Stud",  
  "wire_dia_in": 0.562,  
  "machine":   "MM-14",  
  "station_1": {  
    "description": "Upset / Point Initiation",  
    "die":        "MM-562-D1",  
    "punch":      "MM-562-P1",  
    "punch_pin":  "MM-562-PP1",  
    "spacer":     "MM-562-SP1",  
    "ko_pin":     "MM-562-KO1",  
  },  
  "station_2": {  
    "description": "Cone / Point Forming",  
    "die":        "MM-562-D2",  
    "punch":      "MM-562-P2",  
    "punch_pin":  "MM-562-PP2",  
    "spacer":     "MM-562-SP2",  
    "ko_pin":     "MM-562-KO2",  
  },  
  "station_3": {  
    "description": "Head Form / Point Finish",  
    "die":        "MM-562-D3",  
    "punch":      "MM-562-P3",  
    "punch_pin":  "MM-562-PP3",  
    "spacer":     "MM-562-SP3",  
    "ko_pin":     "MM-562-KO3",  
  },  
  "station_4": {  
    "description": "Final Head Coin / Size",  
    "die":        "MM-562-D4",  
    "punch":      "MM-562-P4",  
    "punch_pin":  "MM-562-PP4",  
    "spacer":     "MM-562-SP4",  
    "ko_pin":     "MM-562-KO4",  
  },  
  "fingers": {  
    "finger_1":   "MM-562-F1",  
    "finger_2":   "MM-562-F2",
```

```

    },
    "notes": (
        "All 4 stations active. "
        "9/16\" and 5/8\" share the same finger pair design — "
        "F1 and F2 are NOT interchangeable with smaller families. "
        "Punch pins PP3 and PP4 are longest in the shop — store vertically."
    ),
},

```

```

# -----
# 5/8" WELD STUD FAMILY (wire dia 0.625")
# -----

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```

"MM-625-WS": {
    "family":    "5/8\" Weld Stud",
    "wire_dia_in": 0.625,
    "machine":   "MM-14",
    "station_1": {
        "description": "Upset / Point Initiation",
        "die":         "MM-625-D1",
        "punch":       "MM-625-P1",
        "punch_pin":   "MM-625-PP1",
        "spacer":      "MM-625-SP1",
        "ko_pin":      "MM-625-KO1",
    },
    "station_2": {
        "description": "Cone / Point Forming",
        "die":         "MM-625-D2",
        "punch":       "MM-625-P2",
        "punch_pin":   "MM-625-PP2",
        "spacer":      "MM-625-SP2",
        "ko_pin":      "MM-625-KO2",
    },
    "station_3": {
        "description": "Head Form / Point Finish",
        "die":         "MM-625-D3",
        "punch":       "MM-625-P3",
        "punch_pin":   "MM-625-PP3",
        "spacer":      "MM-625-SP3",
        "ko_pin":      "MM-625-KO3",
    },
    "station_4": {
        "description": "Final Head Coin / Size",
        "die":         "MM-625-D4",
        "punch":       "MM-625-P4",
    },
}

```

```

        "punch_pin": "MM-625-PP4",
        "spacer": "MM-625-SP4",
        "ko_pin": "MM-625-KO4",
    },
    "fingers": {
        "finger_1": "MM-625-F1",
        "finger_2": "MM-625-F2",
    },
    "notes": (
        "All 4 stations active. "
        "Heaviest wire on MM-14 — run machine at 80% speed until "
        "confirmed good on first 25 pieces. "
        "ST4 die wears fastest of all families — inspect every 40K pieces. "
        "Feed roll pressure will be highest setting in the shop for this family."
    ),
},
}

```

#

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```

# FLAT LIST — every tooling ID across all families
# Use this to seed the tooling_inventory table in the database
#

```

—

```

def get_all_tooling_ids():
    """Returns a flat list of every tooling ID for database seeding."""
    ids = []
    for family_key, family in tooling_lists.items():
        for st in ["station_1", "station_2", "station_3", "station_4"]:
            station = family[st]
            for tool_type in ["die", "punch", "punch_pin", "spacer", "ko_pin"]:
                ids.append({
                    "tool_id": station[tool_type],
                    "tool_type": tool_type,
                    "family": family_key,
                    "station": st,
                    "description": station["description"],
                    "qty_on_hand": 2, # default seed qty
                    "min_qty": 1,
                })

```

```

for finger_key, finger_id in family["fingers"].items():
    ids.append({
        "tool_id":    finger_id,
        "tool_type":  "finger",
        "family":     family_key,
        "station":    finger_key,
        "description": f'{family["family"]} transfer finger',
        "qty_on_hand": 2,
        "min_qty":    1,
    })
return ids

```

```

if __name__ == "__main__":
    ids = get_all_tooling_ids()
    print(f"Total tooling IDs across all families: {len(ids)}")
    print("\nSample — 1/4" family tooling:")
    for item in ids:
        if "250" in item["tool_id"]:
            print(f" {item['tool_id']}<20} {item['tool_type']<12} {item['station']}")

```